Safe Motion Chart - Joint Ranges

Heavy metal 4-DOF arm + gripper | Arduino Uno + PCA9685

File: docs/safe_motion_chart.md

Safe Motion Chart · Joint Ranges for Stable Pick-and-Place

Arm type: Heavy metal 4-DOF + gripper (MG995/MG996R joints)

Controller: Arduino Uno + PCA9685 (I2C, 50 Hz)

Purpose: Keep the arm stable, avoid servo overload and mechanical collision.

Status: **Calibrated on hardware (2026-10-23)** · ranges verified by physical testing.

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1. Channel Map & Calibrated Ranges (HARDWARE VERIFIED)

Joint	Ch	Safe Min	Center	Safe Max	Notes
Base (waist)	**0**	200	375	500	Verified smooth over full range
Shoulder	**1**	200	260	340	260 = vertical; heavy arm - stay compact
Elbow	**2**	200	300	420	300 = vertical/straight
Wrist	**6**	300	375	450	Keep neutral during transport
Wrist rotation	**4**	·	·	·	**SERVO DEAD** - no movement; replace
Gripper	**7**	120 (open)	·	200 (closed)	Verified working; 120-200 only travel
PCA9685 ch3	3	·	·	·	**FAULTY OUTPUT** - do not use

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2. Why the arm tipped / collapsed (root cause)

The arm is **heavy metal** with 2 extra servos at the end effector · large center-of-mass offset. Rotating the base while the arm was **extended** created a tipping moment, and the shoulder could not hold the load.

Rule: fold the arm UPRIGHT before every base rotation.

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3. Pick-and-Place Sequence (implemented in code)

Step	Base	Shoulder	Elbow	Wrist	Gripper	Delay	Purpose
Home	375	260	300	375	120 (open)	2000	Compact, balanced
1. Reach	300	310	250	375	120	800	Lean to object (small)
2. Grip	300	310	250	375	200 (closed)	700	Grasp
3. Lift	300	260	300	375	200	800	Back upright
4. Rotate	·bin	260	300	375	200	600+900	Rotate ONLY while compact
5. Lower	bin	310	250	375	200	700	Down to bin
6. Release	bin	310	250	375	120	700	Open
7. Lift	bin	260	300	375	120	600	Upright
8. Home	375	260	300	375	120	2000	Reset

Bin angles: RED=250, GREEN=375, BLUE=475 (all inside 200-500 safe base range)

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4. Safety Rules (heavy metal arm)

1. **Clamp the base to the table** · mandatory.
2. **Fold upright before rotating** the base (arm compact = balanced).
3. **Sequential joint motion** with 15-25 ms gaps (limits peak current).
4. **Never** command outside calibrated min/max (code clamps automatically).
5. **Power:** XL4016 5-6 V, ·6 A peak, common ground, 2200 µF cap.
6. **Payload** · 180 g; start with · 100 g.
7. **No full-range sweeps** on the assembled arm.
8. **Spot/support** the arm on first runs.

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5. Known Hardware Faults

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| Fault | Symptom | Action |
|---|---|---|
| PCA9685 channel 3 | No output on any servo | Avoid ch3 |
| Wrist rotation servo | Holds but never rotates (any channel, bare shaft too) | **Replace servo** |
| Gripper travel | Only 120-200 ticks move; past that it clamps | Use 120/200 only |
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6. Calibration Procedure (per joint)

1. Power off; run `joint_calibration/joint_calibration.ino`.
2. Sweep in **10-tick steps**, observe for binding/collision.
3. Record safe min/max (keep 10-20 tick margin from hard stops).
4. Update `*_MIN/*_MAX` constants in `pick_and_place.ino` and `integrated_sorting.ino`.
5. Re-run smoothness sweep over the final range.